OPERATIONS / PROPOSED PILOT

Less guesswork at the transfer station.

Test whether a small calibration assistant can reduce manual tuning and positioning exceptions at one material-transfer station - while paying for the time it spends learning.

01 / DECISION REQUEST

Approve discovery and guarded bench validation. Define one station and one product family. Use six weeks as a planning window and **$35,000 as an illustrative allowance**, to be re-estimated after the baseline and fit check. This is not a production rollout request.

Owners: Operations sponsor and Automation Engineering lead, with a station owner, Quality, and Finance. **First comparison:** mechanical guides, a stable fixed recipe, and passive learning. Reject the idea if a simpler option offers better total value.

02 / SAMPLE ECONOMICS - ASSUMPTIONS, NOT ACHIEVED SAVINGS

Annual net benefit	Simple value payback	First-year net value
$46,600	**9.0 months**	**$11,600**

Entered input	Sample assumption
Transfers per year	1,200,000
Exceptions per 1,000 transfers	12 before / 8 target
Incremental value per avoided case	$12
Added time and support per year	$5,000 + $6,000
Pilot and integration, one time	$35,000

Calculation: 4,800 avoided cases x $12 = $57,600 gross annual benefit. Subtract $5,000 added time and $6,000 support for $46,600 annual net benefit. Subtract $35,000 upfront for $11,600 first-year net value.

Downside: a target of 10 exceptions per 1,000 gives $17,800 annual net benefit and 23.6-month payback. With no improvement, annual net value is -$11,000. At these costs, a 12-month payback needs an after rate no higher than about 8.8 per 1,000.

Time saved is capacity, not cash, unless Finance confirms an avoided cost. Avoid double-counting downtime. The simple model excludes taxes, financing, and discounting. All cost and rate inputs are unverified placeholders; no simulation percentage is used as a production rate.

EVIDENCE / VALIDATION PLAN

Earn the next stage with evidence.

The prototype models one-dimensional free sliding with known release speed. It has no live conveyor, PLC, camera, or physical safety integration.

03 / OBSERVED IN SIMULATION

Successful runs / 100	Fixed	Passive	Active	Robust
Unknown fixed friction	83	99	100	27
Friction changes	65	90	89	48
Extra unmodeled drag	88	100	99	24

1,200 MuJoCo runs. A run can use up to six glides; it is not one production transfer. On fixed friction, Active used 3.42 moves and Passive used 2.98. The extra 0.44 moves must earn their cost. No between-strategy significance test was run.

04 / SIX-WEEK PLANNING WINDOW

Phase	Required output
Week 1 / Baseline	Confirm intervention cost, cycle time, product groups, and cheaper alternatives. Agree on the minimum worthwhile effect.
Week 2 / Guarded bench	Measure release speed and stopping error. Check model fit, fallback, and a safe test envelope.
Weeks 3-4 / Controlled trial	Start with operator-reviewed recommendations. Use approved windows and matched or randomized product/shift blocks.
Weeks 5-6 / Decision	Review all exceptions, added time, overrides, damage, uncertainty, and cost. Stop, extend, or separately approve a next stage.

05 / PROPOSED ACCEPTANCE AND STOP GATES

- **Effect:** at least 20% lower intervention rate in matched groups, with pre-agreed uncertainty analysis.
- **Guardrails:** no more than 2% throughput loss; no worsening of damage, jams, or stop events.
- **Value:** measured costs meet the agreed payback target, including probe time and support.
- **Ownership:** approved equipment envelope, rollback, setting changes, and incident handling.

Stop or redesign if the model cannot explain real motion, release speed is unreliable, a simpler alternative wins, or added time removes the benefit. A positive calculator result never approves deployment.

Size the trial from measured baseline rarity, the minimum useful effect, and product/shift clustering. Six weeks is not a sample-size guarantee. Pricing and willingness to pay remain unvalidated.

Industry context only: Dorner: product-transfer methods; Siemens: virtual commissioning for material handling. These sources support the problem domain and test workflow, not this prototype's performance.